

Problem Set 11

Problem 10.2.3:

- (1) Since $H = H_x + H_y + H_z$ where $H_x = P_x^2/(2m) + X^2(m\omega^2/2)$, etc., the problem separates into 3 copies of the 1-dimensional harmonic oscillator. So the eigenstates of H are the tensor products of the eigenstates of H_x, H_y, H_z :

$$|E\rangle = |E_x\rangle|E_y\rangle|E_z\rangle \quad \text{with } E_x = (n_x + \frac{1}{2})\hbar\omega, \quad n_x \in \{0, 1, 2, \dots\}, \quad \text{etc.}$$

Thus the spectrum is

$$E = E_x + E_y + E_z = (n_x + n_y + n_z + \frac{3}{2})\hbar\omega, \quad n_x, n_y, n_z \in \{0, 1, 2, \dots\}.$$

- (2) In the position basis for the 1-d harmonic oscillator $|E_n\rangle \rightarrow \psi_n(x)$ with parity $(-)^n$. So

$$|E\rangle = |E_{n_x}\rangle|E_{n_y}\rangle|E_{n_z}\rangle \longrightarrow \psi_{n_x}(x)\psi_{n_y}(y)\psi_{n_z}(z) := \psi_{\vec{n}}(\vec{x})$$

has parity under $\vec{x} \rightarrow -\vec{x}$ of $(-)^{n_x}(-)^{n_y}(-)^{n_z} = (-)^{n_x+n_y+n_z} = (-)^n$.

Level 0: $n = 0$, therefore $n_x = n_y = n_z = 0$, so

$$\psi_{\vec{0}}(\vec{x}) = \psi_0(x)\psi_0(y)\psi_0(z) = \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} e^{-m\omega(x^2+y^2+z^2)/(2\hbar)} = \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} e^{-m\omega r^2/(2\hbar)}$$

since $r^2 = x^2 + y^2 + z^2$ in spherical coordinates.

Level 1: $n = 1$, so there are three possibilities $\vec{n} = (1, 0, 0)$ or $(0, 1, 0)$ or $(0, 0, 1)$, so

$$\psi_{(1,0,0)} = \psi_1(x)\psi_0(y)\psi_0(z) = \frac{1}{\sqrt{2}} \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} e^{-m\omega(x^2+y^2+z^2)/(2\hbar)} \cdot 2 \left(\frac{m\omega}{\hbar}\right)^{1/2} x,$$

$$\psi_{(0,1,0)} = \psi_0(x)\psi_1(y)\psi_0(z) = \frac{1}{\sqrt{2}} \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} e^{-m\omega(x^2+y^2+z^2)/(2\hbar)} \cdot 2 \left(\frac{m\omega}{\hbar}\right)^{1/2} y,$$

$$\psi_{(0,0,1)} = \psi_0(x)\psi_0(y)\psi_1(z) = \frac{1}{\sqrt{2}} \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} e^{-m\omega(x^2+y^2+z^2)/(2\hbar)} \cdot 2 \left(\frac{m\omega}{\hbar}\right)^{1/2} z.$$

Since in spherical coordinates $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$, we then have

$$\psi_{(1,0,0)} = \sqrt{2\pi} \left(\frac{m\omega}{\pi\hbar}\right)^{5/4} r \sin \theta \cos \phi e^{-m\omega r^2/(2\hbar)},$$

$$\psi_{(0,1,0)} = \sqrt{2\pi} \left(\frac{m\omega}{\pi\hbar}\right)^{5/4} r \sin \theta \sin \phi e^{-m\omega r^2/(2\hbar)},$$

$$\psi_{(0,0,1)} = \sqrt{2\pi} \left(\frac{m\omega}{\pi\hbar}\right)^{5/4} r \cos \theta e^{-m\omega r^2/(2\hbar)}.$$

Level n degeneracy: Since $E_n = (n + \frac{3}{2})\hbar\omega$ with $n = n_x + n_y + n_z$ where $n_{x,y,z}$ are non-negative integers, we only have to count the number of ways of choosing (n_x, n_y, n_z) to add up to n . For a given n , there are $n + 1$ different possible values of $n_x = (0, 1, \dots, n)$, and for given n and n_x , there are $n - n_x + 1$ different values of n_y . n_z is then fixed. So there are a total of

$$\begin{aligned} \sum_{n_x=0}^n (n - n_x + 1) &= (n + 1) \left(\sum_{n_x=0}^n 1 \right) - \left(\sum_{n_x=0}^n n_x \right) = (n + 1)^2 - \frac{n(n + 1)}{2} \\ &= \frac{(n + 1)(n + 2)}{2} \end{aligned}$$

different ways of choosing (n_x, n_y, n_z) .

Problem 10.3.1: Since we have identical bosons, we need to make a symmetrized state vector out of the tensor product of $|\phi\rangle$ and $|\psi\rangle$:

$$|\phi\psi, S\rangle = \alpha (|\phi\rangle|\psi\rangle + |\psi\rangle|\phi\rangle) \quad \text{for some number } \alpha.$$

To find α , normalize:

$$\begin{aligned} 1 &= \langle\phi\psi, S|\phi\psi, S\rangle = |\alpha|^2 (\langle\phi|\langle\psi| + \langle\psi|\langle\phi|) (|\phi\rangle|\psi\rangle + |\psi\rangle|\phi\rangle) \\ &= |\alpha|^2 (\langle\phi\psi|\phi\psi\rangle + \langle\phi\psi|\psi\phi\rangle + \langle\psi\phi|\phi\psi\rangle + \langle\psi\phi|\psi\phi\rangle) \\ &= |\alpha|^2 (\langle\phi|\phi\rangle\langle\psi|\psi\rangle + \langle\phi|\psi\rangle\langle\psi|\phi\rangle + \langle\psi|\phi\rangle\langle\phi|\psi\rangle + \langle\psi|\psi\rangle\langle\phi|\phi\rangle) \\ &= 2|\alpha|^2 (\langle\phi|\phi\rangle\langle\psi|\psi\rangle + |\langle\phi|\psi\rangle|^2). \end{aligned}$$

If $|\phi\rangle$ and $|\psi\rangle$ are normalized, $\langle\phi|\phi\rangle = \langle\psi|\psi\rangle = 1$, and we have $1 = 2|\alpha|^2(1 + |\langle\phi|\psi\rangle|^2)$. So the normalized bosonic state is

$$|\phi\psi, S\rangle = \frac{1}{\sqrt{2(1 + |\langle\phi|\psi\rangle|^2)}} (|\phi\rangle|\psi\rangle + |\psi\rangle|\phi\rangle).$$

Problem 10.3.6: Say we have a collection of particles $B_1, B_2, \dots, B_b$ and $F_1, F_2, \dots, F_f$ where the B 's are bosons and the F 's are fermions (not necessarily identical), and they have coordinates $(x_1, \dots, x_b, y_1, \dots, y_f)$ respectively. Then the wavefunction

$$\psi(x_1, \dots, x_b, y_1, \dots, y_f; \tilde{x}_1, \dots, \tilde{x}_b, \tilde{y}_1, \dots, \tilde{y}_f)$$

for *two* identical such collections must be symmetric under interchanges $x_i \leftrightarrow \tilde{x}_i$ (bosons, $\psi \rightarrow +\psi$) and antisymmetric under interchanges $y_i \leftrightarrow \tilde{y}_j$ (fermions, $\psi \rightarrow -\psi$). So, if we interchange all the $(x_i, y_j) \leftrightarrow (\tilde{x}_i, \tilde{y}_j)$ we must have

$$\psi(x_1, \dots, x_b, y_1, \dots, y_f; \tilde{x}_1, \dots, \tilde{x}_b, \tilde{y}_1, \dots, \tilde{y}_f) = (-)^f \psi(\tilde{x}_1, \dots, \tilde{x}_b, \tilde{y}_1, \dots, \tilde{y}_f; x_1, \dots, x_b, y_1, \dots, y_f).$$

This means that under interchange of the two identical *sets* of particles, the wavefunction gets a $(-)^f$ sign. Thus if the number of fermions, f , in each set is even, $(-)^f = +1$, and the sets act as bosons. If f is odd, $(-)^f = -1$, and the sets act as fermions.

[A puzzle: If even numbers of fermions behave as bosons, do they not satisfy the exclusion principle? To answer this, consider a 2-fermion system (of non-identical fermions) with wavefunction

$$|\psi_{2f}\rangle = \sum_{n,m} c_{n,m} |n, m\rangle,$$

where the index n in $|n, m\rangle$ labels the (1-particle) states of the first fermion, and m labels the states of the second fermion. Now, a second (identical) copy of this system, $|\psi_{2\tilde{f}}\rangle$, is made. Can they both be in the same state (together)?

$$\begin{aligned} |\psi_{2f+2\tilde{f}}\rangle &= |\psi_{2f}\rangle \otimes |\psi_{2\tilde{f}}\rangle \pm \dots \\ &\propto \sum_{n,m,\tilde{n},\tilde{m}} c_{nm} c_{\tilde{n}\tilde{m}} (|nm\rangle |\tilde{n}\tilde{m}\rangle - |n\tilde{m}\rangle |\tilde{n}m\rangle - |\tilde{n}m\rangle |n\tilde{m}\rangle + |\tilde{n}\tilde{m}\rangle |nm\rangle) \\ &\propto \sum_{n,\tilde{n},m,\tilde{m}} (c_{nm} c_{\tilde{n}\tilde{m}} - c_{\tilde{n}m} c_{n\tilde{m}} - c_{n\tilde{m}} c_{\tilde{n}m} + c_{\tilde{n}\tilde{m}} c_{nm}) |nm, \tilde{n}\tilde{m}\rangle \\ &\propto 2 \sum_{n,\tilde{n},m,\tilde{m}} (c_{nm} c_{\tilde{n}\tilde{m}} - c_{\tilde{n}m} c_{n\tilde{m}}) |nm, \tilde{n}\tilde{m}\rangle \end{aligned}$$

which need *not* vanish, in general. So, indeed, there is no exclusion principle! (In the first line the $\pm \dots$ stands for antisymmetrization separately under interchange of the first fermion states in each system as well as under interchange of the second fermion states.) However, notice that in the final sum that any terms with $n = \tilde{n}$ or $m = \tilde{m}$ vanish. Thus, in particular, if there were only one internal fermion state that, say, the first fermion could be in, then the sum would vanish, and there *would* be an exclusion principle. This statement generalizes as follows: if a component fermion of a multi-fermion system is in a superposition of only N states, then antisymmetrization will forbid having more than N multi-fermion identical systems, even if those systems act as bosons.]